Reproducible Research Repository

[redacted]

*First Submission: RR_WLD_2026_*[redacted]

[redacted]

reproducibility@worldbank.org
April 8, 2026

This review verifies the reproducibility of the exhibits included in the paper "[redacted]".

Contents in this review:

1. Main findings
2. List of exhibits and reproducibility status
3. Reproduction Environment

Main findings

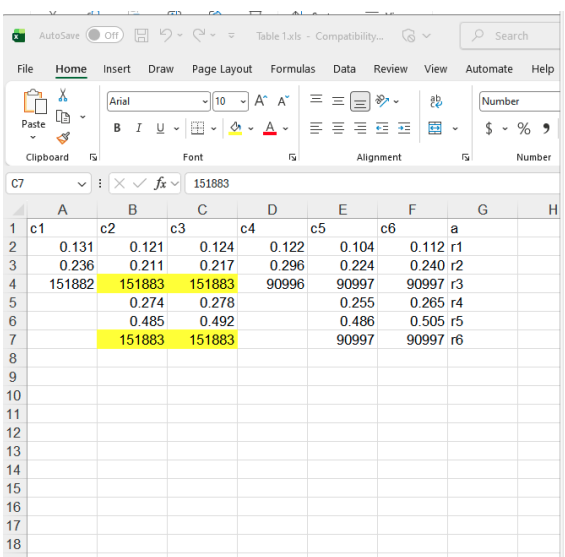
- **The exhibits in the paper did not meet the requirements for computational reproducibility, with 5 exhibits failing to reproduce accurately.**
- The code was successfully executed on a new computer after:
  1. Updating the working directory in line 10 of the do-file "main.do", and running it.
  2. Opening the R project "inequality_analysis.Rproj"
  3. Restoring the environment by running renv::restore() and following the prompts.
  4. Opening the main.R and running the code.
- The code takes approximately 2 days to run.
- We conducted our reproducibility analysis based on the paper shared by the authors.
- **Verification Process and Data Handling**:
  - The reproducibility package relies on 3 types of data: open, accessible (public but not included in the reproducibility package), and restricted data (available only from the authors).
  - Open data is included in the public reproducibility package.
  - Reviewers used forthcoming/accessible/restricted data provided directly by the authors to conduct the reproducibility verification, and this is not included in the public reproducibility package.
  - *data_hash_report.csv* lists the SHA256 hashes of all files in the Data folder to support data integrity checks. Users who acquire the forthcoming/accessible/limited-access/restricted data can use this file to verify that the data has not been altered.

- **Reproducibility Summary**:
  - **Data**: Some data is restricted and has not been included in the reproducibility package. For more details, please refer to the README file.
  - **Code**: All code files (from cleaning to analysis) are included in the reproducibility package.
  - **Outputs**: All outputs are generated by code included in the reproducibility package.
  - **Dependencies environment**: The reviewers created a new environment using the latest versions of dependencies available at the moment of the review.

List of exhibits and reproducibility status

Results in the Main Section of the Paper

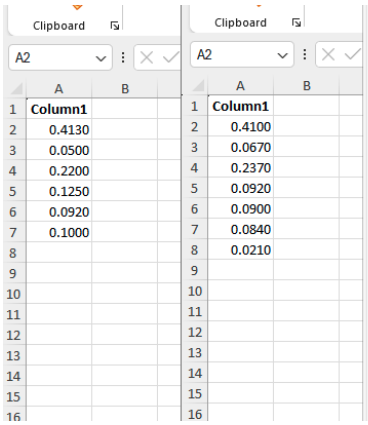
- **Figure 1** Reproduced.
- **Figure 2** Reproduced.
- **Figure 3** Reproduced.
- **Figure 4** Reproduced.
- **Table 1** Does not reproduce. See below:



	A	B	C	D	E	F	G
1	c1	c2	c3	c4	c5	c6	a
2	0.131	0.121	0.124	0.122	0.104	0.112	r1
3	0.236	0.211	0.217	0.296	0.224	0.240	r2
4	151882	151883	151883	90996	90997	90997	r3
5		0.274	0.278		0.255	0.265	r4
6		0.485	0.492		0.486	0.505	r5
7		151883	151883		90997	90997	r6

Panel A: MLD	Outcome: Per capita Income			Outcome: Labor Income		
	Parametric	Conditional Trees	Random Forest	Parametric	Conditional Trees	Random Forest
IO	0.131	0.121	0.124	0.122	0.104	0.112
IOr	23.6	21.1	21.7	29.6	22.4	23.9
Obs	151,882	151,882	151,882	90,996	90,997	90,997
Panel B: Gini						
	Outcome: Per capita Income			Outcome: Labor Income		
	Parametric	Conditional Trees	Random Forest	Parametric	Conditional Trees	Random Forest
IO	NA	0.274	0.278	NA	0.255	0.265
IOr	NA	48.5	49.2	NA	48.6	50.5
Obs	NA	151,882	151,882	NA	90,997	90,997

- **Table 2** Does not reproduce. The row "Place of birth" is not exported in the code output, so we were unable to verify it. In addition, the values in the last two columns differ from the manuscript table, and the differences exceed 0.01. See below:

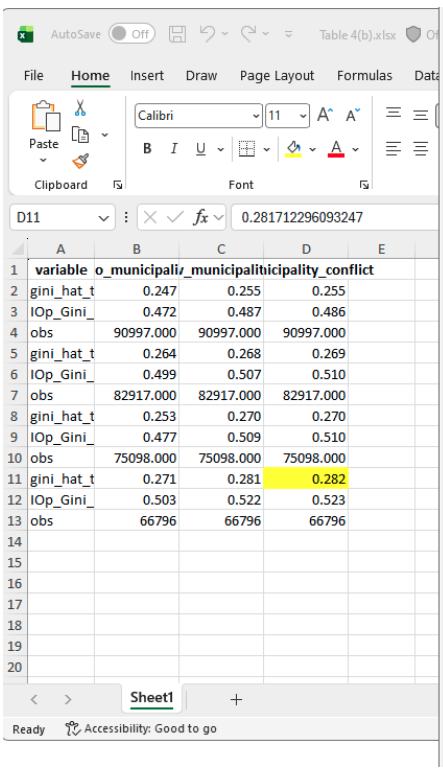


	Column1
1	0.4130
2	0.0500
3	0.2200
4	0.1250
5	0.0920
6	0.1000

	Column1
1	0.4100
2	0.0670
3	0.2370
4	0.0920
5	0.0900
6	0.0840
7	0.0210

	Shapley (under parametric approach using MLD) %		Contribution (under random forest) %	
	Per capita Income	Labor Income	Per capita Income	Labor Income
Parents Education	**54.7**	**53.6**	**41.2**	**41.6**
Ethnicity	**6.5**	**9.2**	**5.2**	**6.4**
Place of birth	**38.8**	**35.7**	**53.6**	**50.1**
Department of birth	2.7	2	22.5	23.6
Municipality category of birth	17.4	17.1	12.1	8.8
Rural/Urban Municipality	10.3	11	9.2	9.2
Presence of Internal Armed Conflict in Municipality when Born	8.4	5.6	9.8	8.5
Sex	**na**	**1.6**	**na**	**1.9**

- **Table 3** Reproduced.
- **Table 4** Does not reproduce. One value does not reproduce. See below:



	A	B	C	D
1	variable	o_municipali	_municipaliti	icipality_conflict
2	gini_hat_t	0.247	0.255	0.255
3	IOp_Gini_	0.472	0.487	0.486
4	obs	90997.000	90997.000	90997.000
5	gini_hat_t	0.264	0.268	0.269
6	IOp_Gini_	0.499	0.507	0.510
7	obs	82917.000	82917.000	82917.000
8	gini_hat_t	0.253	0.270	0.270
9	IOp_Gini_	0.477	0.509	0.510
10	obs	75098.000	75098.000	75098.000
11	gini_hat_t	0.271	0.281	0.282
12	IOp_Gini_	0.503	0.522	0.523
13	obs	66796	66796	66796

Specification type/key variable		Per capita income: Without municipality variables	Per capita income: With municipality variables	Per capita income: Municipality + internal armed conflict	Labor income: Without municipality variables	Labor income: With municipality variables	Labor income: Municipality + internal armed conflict
S1: Education imputed, one variable for both parents	IO	0.255	0.264	0.274	0.247	0.255	0.255
	IOr	45.0%	46.7%	48.5%	47.2%	48.7%	48.6%
	N		151,883			90,997	
S2: Education no imputed, one variable for both parents	IO	0.264	0.271	0.278	0.264	0.268	0.269
	IOr	46.2%	47.5%	48.8%	49.9%	50.7%	51.0%
	N		138,262			82,917	
S3: Education imputed, father and mother separated	IO	0.265	0.274	0.281	0.253	0.270	0.270
	IOr	46.3%	47.8%	49.1%	47.7%	50.9%	51.0%
	N		124,676			75,098	
S4: Education no imputed, father and mother separated	IO	0.270	0.281	0.287	0.271	0.281	0.126
	IOr	46.6%	48.5%	49.5%	50.3%	52.2%	52.3%
	N		111,812			66,796	

- **Table 5** Reproduced.

Results in the Annex

For the Appendix, we did not review every exhibit. Instead, we randomly selected 10 exhibits from the remaining datasets to assess the appendix. Our review was based on those 10 exhibits. Since they were chosen randomly, we are operating under the assumption that if all randomly selected exhibits are reproducible, then the rest should be as well. The seed used to generate the random selection was

961021, the Stata version used was 18, and the exhibits selected were: Tables A2, A3, A4, A6, and Figures A1, A2, A3, A6, A7, A8

- **Table A2** Does not reproduce. One value does not reproduce. See below:

E11 | 0.14926378428936

	A	B	C	D	E	F	G
1	c1	c2	c3	c4	c5	c6	a
2	0.095	0.111	0.131	0.091	0.107	0.122	r1
3	0.171	0.200	0.236	0.223	0.261	0.296	r2
4	151882.000	151882.000	151882.000	90996.000	90996.000	90996.000	r3
5	0.106	0.122	0.135	0.107	0.123	0.133	r4
6	0.186	0.215	0.238	0.252	0.291	0.313	r5
7	138262.000	138262.000	138262.000	82916.000	82916.000	82916.000	r6
8	0.113	0.128	0.144	0.112	0.127	0.139	r7
9	0.197	0.222	0.251	0.260	0.296	0.323	r8
10	124675.000	124675.000	124675.000	75097.000	75097.000	75097.000	r9
11	0.123	0.140	0.157	0.131	0.149	0.157	r10
12	0.209	0.237	0.266	0.292	0.333	0.350	r11
13	111812.000	111812.000	111812.000	66795.000	66795.000	66795.000	r12

		Per capita income			Labor income		
Specification type/key variable		Without municipality variables (1)	With municipality variables (2)	Municipality + internal armed conflict (3)	Without municipality variables (4)	With municipality variables (5)	Municipality + internal armed conflict (6)
S1: Education imputed, one variable for both parents	IO	0.095	0.111	0.131	0.091	0.107	0.122
	IOr	17.1%	20.0%	23.6%	22.3%	26.1%	29.6%
	N			151,882			90,996
S2: Education no imputed, one variable for both parents	IO	0.106	0.122	0.135	0.107	0.123	0.133
	IOr	18.6%	21.5%	23.8%	25.2%	29.1%	31.3%
	N			138,262			82,916
S3: Education imputed, father and mother separated	IO	0.113	0.128	0.144	0.112	0.127	0.139
	IOr	19.7%	22.2%	25.1%	26.0%	29.6%	32.3%
	N			124,675			75,097
S4: Education no imputed, father and mother separated	IO	0.123	0.140	0.157	0.131	0.162	0.157
	IOr	20.9%	23.7%	26.6%	29.2%	33.3%	35.0%
	N			111,812			66,795

- **Table A3** Reproduced.
- **Table A4** Reproduced.
- **Table A6** Reproduced.
- **Figure A1** Results reproduced. Figure was not exported by code but was verified from the results window.
- **Figure A2** Could not be verified. Figure not found in code outputs.
- **Figure A3** Reproduced. Verified against Figure A2.png and Figure A3.png.
- **Figure A6** Reproduced.
- **Figure A7** Reproduced.
- **Figure A8** Reproduced.

Reproduction Environment

Paper exhibits were reproduced on a computer with the following specifications:

- OS: Windows 11 Enterprise
- Processor: INTEL(R) XEON(R) PLATINUM 8562Y+ 2.80 GHz (2 processors)
- Memory available: 128 GB
- Software version: Stata 18.0 MP; R 4.3.2